- *Ongoing maintenance* costs relate to maintaining compliance with updated and evolving standards and keeping your systems aligned with the latest protocols.

5.5.3 A practical assessment framework

A pragmatic approach to standards adoption involves two key questions. Answering them will help you prioritize and decide which standards are truly important for your digital twin journey.

1. *Does it solve a problem?* This is the most critical question. A standard should be adopted because it provides a clear solution to a specific challenge you are facing. For example:
    - *Interoperability* - if your digital twin needs to seamlessly integrate with other systems, such as a Building Information Modeling (BIM) platform, adopting a standard like IFC (Industry Foundation Classes) is essential. It provides a common language for data exchange, solving the problem of data silos.
    - *Data integrity* - if you need to ensure the accuracy and reliability of data from various sensors and devices, a communication protocol standard like *OPC-UA (Open Platform Communications Unified Architecture)* can solve this by providing a robust, secure, and standardized way to transfer data.
2. *Is it a widely adopted and relevant standard?* Sometimes, a standard's value isn't just about solving a current problem but about its future relevance and industry-wide acceptance. A standard that is widely adopted is more likely to be supported by a larger ecosystem of tools, vendors, and experts. When considering adoption and relevance, consdier:
    - *Industry consensus* - is the standard backed by a reputable consortium or a leading industry body? Look for standards that have broad support, as this indicates they are likely to remain relevant.
    - *Future-proofing* - adopting a well-established and evolving standard (like an ISO standard for digital twins) can future-proof your system, making it easier to integrate with future technologies and ensuring its long-term viability.